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PLANT SEARCHER

INTRODUCTION

My web project is a plant searcher that allows users to upload photos of a plant via a drag-and-drop box. The application then displays a card with the identified plant and other similar plants, each linking to a dedicated page with extended information.

This project integrates two APIs:

- **PlantNet:** Requires a free API key. I used it to send images and identify the plant or possible matches.

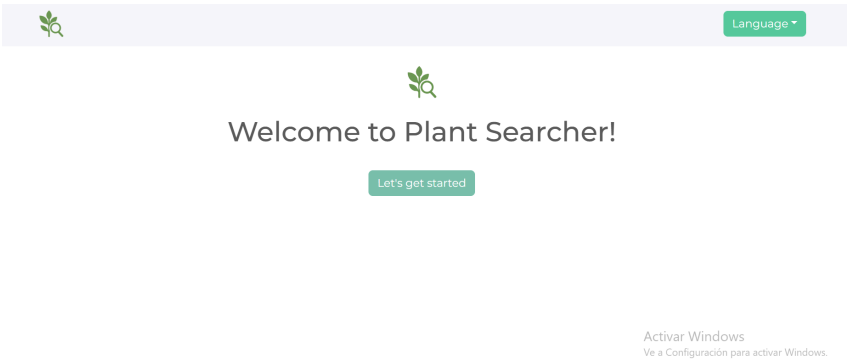
- **Wikipedia:** Used to retrieve detailed information about the plants, including images, descriptions, and external links.

In addition to the APIs, the project uses several libraries and tools to enhance functionality and user experience:

- **React Router**: For managing navigation between pages.
- **React Dropzone:** For drag-and-drop image uploads.
- **i18n (react-i18next):** To provide multilingual support.
- **Zustand:** For global state management
- **Bootstrap & Bootswatch:** For responsive design and theming.

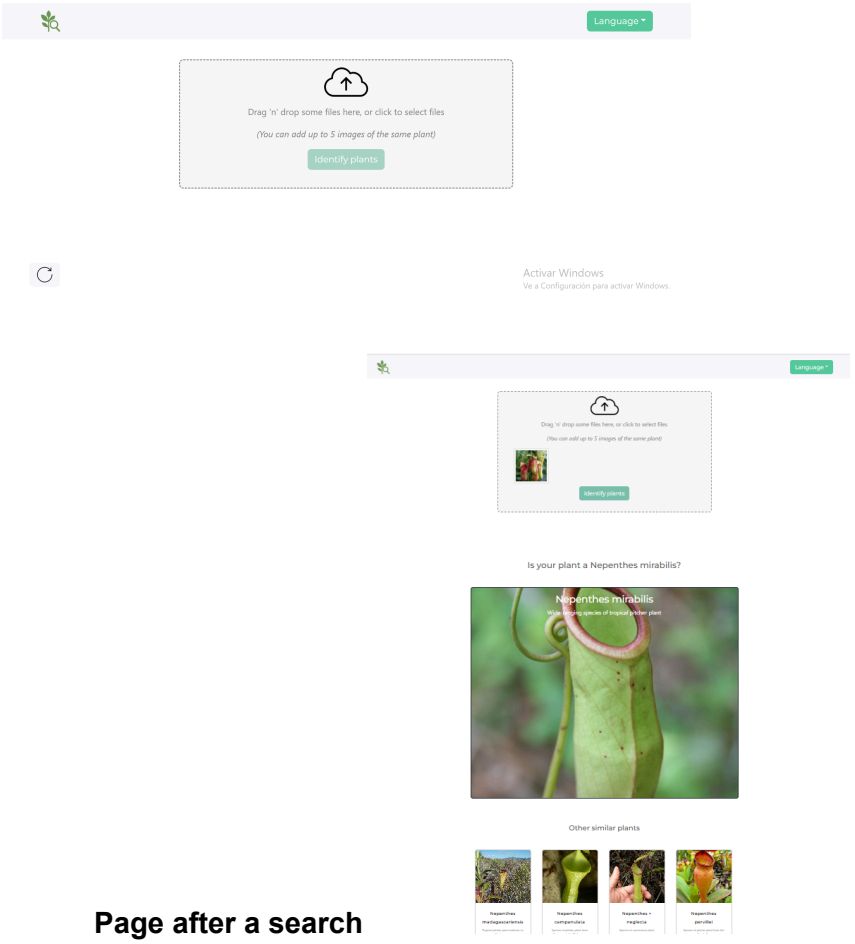
This website uses three main pages to navigate:

1. **The front page:** Landing page with a greeting, logo, and access to the search feature.



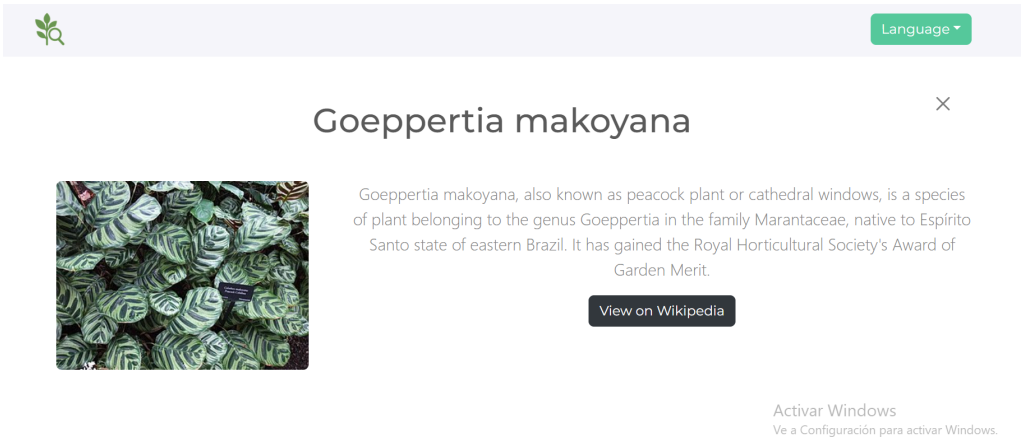
2. **Searcher:** Interactive search page where users can upload an image to identify plants and view results with additional information.

Page before a search



Page after a search

3. **Plant page:** plant detail pages showing the identified plant and information from Wikipedia like a representative image, a brief description of the plant and a link to the Wikipedia page where the information can be found.



OTHER COMPONENTS AND IMPLEMENTATIONS

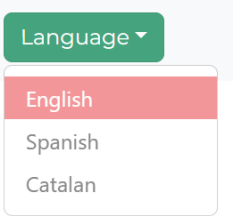
Page navigation

A navigation system that allows users to move between the main page, searcher, and individual plant pages. It ensures a consistent and intuitive browsing experience across the app.

```
<Routes>
  <Route path="/" element={<App />} />
  <Route path="/searcher" element={<Searcher />} />
  <Route path="/plant/:plantName" element={<Plant />} />
</Routes>
```

Multilingual support

The project uses i18n (react-i18next) to provide multilingual support, allowing the interface and content to adapt to English, Spanish, and Catalan. The selected language persists across pages and reloads, including in Wikipedia API requests. If a description is not available in Spanish or Catalan, English is used as a fallback, ensuring that at least an English description is always displayed.



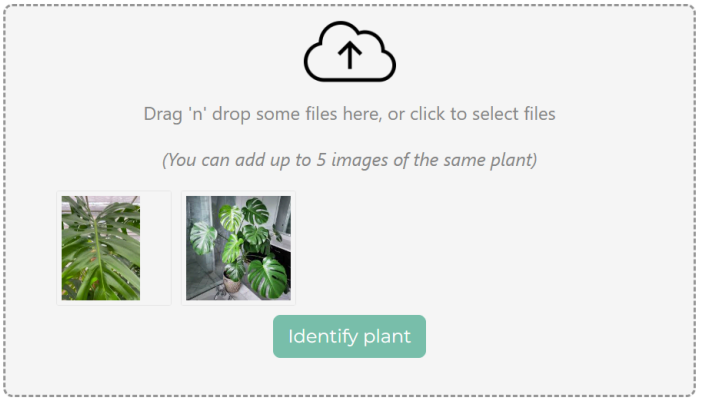
Navbar logo

Clicking the navbar logo takes the user back to the home page, providing quick and intuitive navigation.



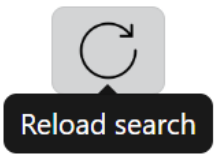
Drag-and-Drop

Multiple images can be selected from the computer or dragged into the drop zone, with previews displayed before doing the search.

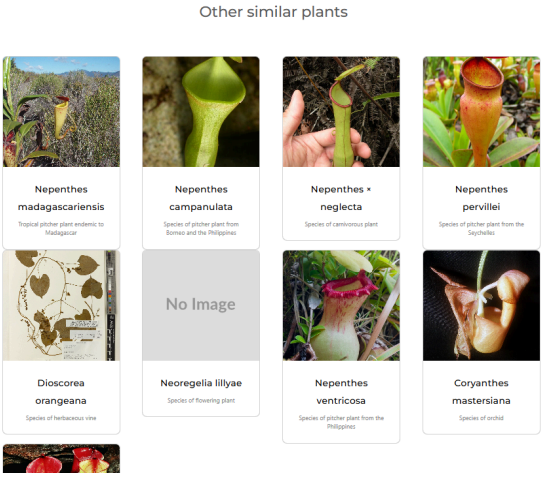


Persistent Search Results

The search results are maintained across pages and even after refreshing the browser, thanks to Zustand for global state management. A new search is only triggered when the dedicated reload button is pressed, ensuring that previous results remain visible until explicitly refreshed.



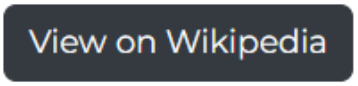
Similar Plants
For each suggested plant, a card displays an image and a brief description. Clicking on a card navigates to a dedicated page with more detailed information about that plant. Additionally, if a plant does not have an associated image, a default placeholder image is displayed.



Return to Searcher
Each plant page includes a close (“×”) button that allows returning to the searcher page quickly and intuitively.



View on Wikipedia
Each plant page includes a button that links directly to the corresponding Wikipedia page used as the source. The link automatically adapts to the selected language, directing to the appropriate version of Wikipedia.



CONCLUSION AND CHALLENGES

During the development of the project, I encountered some challenges, particularly with handling images across multiple drop zones, which I was ultimately unable to fully resolve. Even so, the web app turned out to be really useful, intuitive, and pleasant to look at, making it easy and enjoyable to search for and explore plants.